

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250280672

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

MA; Liang

DISPLAY PANELS

Abstract

The present application provides a display panel. The display panel includes: a photosensitive device, where the photosensitive device includes a switching element, a first photosensitive element, and a second photosensitive element. The first photosensitive element includes a first electrode, a first photosensitive portion, and a first protection electrode. The second photosensitive element includes a second electrode, a second photosensitive portion, and a second protective electrode. The switching element includes a source and a drain. The first electrode is connected to the second electrode, and the first photosensitive portion and the second photosensitive portion are both connected to either the source or the drain.

Inventors: MA; Liang (Wuhan, Hubei, CN)

Applicant: Wuhan China Star Optoelectronics Semiconductor Display Technology Co., Ltd. (Wuhan, Hubei, CN)

Family ID: 86487759

Appl. No.: 18/556672

Filed (or PCT Filed): June 29, 2023

PCT No.: PCT/CN2023/103595

Foreign Application Priority Data

CN 202211478158.0

Nov. 23, 2022

Publication Classification

Int. Cl.: H10K59/131 (20230101); H10K39/34 (20230101)

U.S. Cl.:

Background/Summary

TECHNICAL FIELD

[0001] The present application relates to the field of display technologies, and more particularly to display panels.

BACKGROUND

[0002] A Fingerprint recognition has become a function that most of display terminals such as a mobile phone, a tablet computer, a notebook computer, and the like are equipped with. Currently, the fingerprint recognition of the display device is gradually changing from capacitive fingerprint recognition to optical fingerprint recognition. The optical fingerprint recognition is to image a user's fingerprint by refraction and reflection of light, and then to recognize a fingerprint feature by an image recognition method. The optical fingerprint recognition has characteristics of higher imaging resolution, easies image recognition, and the like, and can be arranged under a display screen to form an underscreen fingerprint recognition.

[0003] A conventional photosensitive device detects a light using a photosensitive element that is a semiconductor device converting a received light signal into an electrical signal. However, with improvement of the pixel resolution, an in-plane light emitting unit may squeeze the area of the photosensitive element, resulting in a lower amount of the photoelectric signal and affecting detection sensitivity of the fingerprint recognition.

SUMMARY

[0004] The present application provides a display panel to solve the technical problems that the area of the photosensitive element is smaller, resulting in a lower amount of the photoelectric signal, and affecting the detection sensitivity.

[0005] The present application provides a display panel, including: at least one photosensitive device, including a switching element, a first photosensitive element, and a second photosensitive element; where the display panel further includes:

[0006] a substrate;

[0007] a first conductive layer disposed on the substrate and including a first electrode and a second electrode connected to the first electrode;

[0008] a photosensitive layer disposed on one side of the first conductive layer away from the substrate and including a first photosensitive portion corresponding to the first electrode and a second photosensitive portion corresponding to the second electrode, where the first photosensitive portion and the second photosensitive portion are disposed at intervals, and the photosensitive layer further includes a metal oxide semiconductor;

[0009] an insulating layer disposed between the first conductive layer and the photosensitive layer;

[0010] a second conductive layer disposed on one side of the photosensitive layer away from the substrate and including a first protection electrode and a second protection electrode, where the first protection electrode is electrically connected to the first photosensitive portion, and the second protection electrode is electrically connected to the second photosensitive portion; and

[0011] a third conductive layer disposed on one side of the second conductive layer away from the substrate and including a source and a drain of the switching element, where the first protection electrode and the second protection electrode are both electrically connected to the source or the drain of the switching element;

[0012] where the first photosensitive element includes the first electrode, the first photosensitive portion, and the first protection electrode, and the second photosensitive element includes the second electrode, the second photosensitive portion, and the second protective electrode.

[0013] Optionally, in some embodiments of the present application, the display panel further includes:

[0014] a first active layer disposed between the substrate and the photosensitive layer and including a first active portion of the switching element, where the source and the drain of the switching element are both electrically connected to the first active portion, and the first active portion includes either a polysilicon semiconductor or a polysilicon semiconductor.

[0015] Optionally, in some embodiments of the present application, the photosensitive device further includes: storage capacitor;

[0016] the first conductive layer further includes a first capacitor electrode of the storage capacitor, where the first capacitor electrode is electrically connected to both the first electrode and the second electrode; and

[0017] the third conductive layer further includes a second capacitor electrode of the storage capacitor, where the second capacitor electrode is electrically connected to both the first protection electrode and the second protection electrode, and at least partially overlaps the first capacitor electrode.

[0018] Optionally, in some embodiments of the present application, the display panel further includes: a fourth conductive layer including a conductive portion, where an orthographic projection of the conductive portion on the substrate at least partially overlaps an orthographic projection of at least one of the first electrode and the second electrode on the substrate.

[0019] Optionally, in some embodiments of the present application, the metal oxide semiconductor of the photosensitive layer has an electron mobility greater than or equal to $10 \text{ cm}^2/\text{Vs}$, and the insulating layer has a thickness of 5 nm to 15 nm.

[0020] Optionally, in some embodiments of the present application, the display panel further includes: a pixel driving circuit including both a first driving transistor and a second driving transistor;

[0021] the display panel further includes:

[0022] a second active layer disposed between the substrate and the photosensitive layer and including a second active portion of the first driving transistor, where the second active portion includes a polysilicon semiconductor;

[0023] a third active layer disposed between the second active layer and the photosensitive layer and including a third active portion of the second driving transistor, where the third active portion includes a metal oxide semiconductor; and

[0024] a conductive layer disposed between the second active layer and the third active layer and including a second gate of the first driving transistor, where the second gate at least partially overlaps the second active portion;

[0025] where the second conductive layer further includes a third gate of the second driving transistor, the third gate at least partially overlapping the third active portion.

[0026] Optionally, in some embodiments of the present application, the first active portion includes a polysilicon semiconductor;

[0027] the second active layer further includes the first active portion, and the conductive layer further includes a first gate of the switching element, the first gate at least partially overlapping the first active portion.

[0028] Optionally, in some embodiments of the present application, the first active portion includes a metal oxide semiconductor.

[0029] the third active layer further includes the first active portion, the conductive layer further includes a fourth gate of the switching element, and the second conductive layer further includes a first gate of the switching element, where the fourth gate at least partially overlaps the first active portion, and the first gate at least partially overlaps the first active portion.

[0030] Optionally, in some embodiments of the present application, the display panel further includes: a plurality of light emitting units arranged in a plurality of rows in a first direction and

arranged in a plurality of columns in a second direction;

[0031] where a plurality of intersecting regions are formed between the plurality of rows of light emitting units and the plurality of columns of light emitting units, and each of the intersecting regions is provided with at most one first photosensitive element or one second photosensitive elements.

[0032] Optionally, in some embodiments of the present application, the first photosensitive element and the second photosensitive element are respectively located in two adjacent ones of the intersecting regions located on the same side of the same light emitting unit.

[0033] Optionally, in some embodiments of the present application, the display panel further includes: a plurality of the photosensitive devices;

[0034] where a plurality of the first photosensitive elements and a plurality of the second photosensitive elements in the same row are alternately arranged, a plurality of the first photosensitive elements and a plurality of the second photosensitive elements in the same column are alternately arranged, and each of the first photosensitive elements is connected in parallel with one of the second photosensitive elements, which is located in the same row and in the same side as the first photosensitive element and adjacent to the first photosensitive element, or each of the first photosensitive elements is connected in parallel with one of the second photosensitive elements, which is located in the same column and in the same side as the first photosensitive element and adjacent to the first photosensitive element; or

[0035] a plurality of the first photosensitive elements and a plurality of the second photosensitive elements in the same row are alternately arranged, the first photosensitive elements or the second photosensitive elements are in the same column, and each of the first photosensitive elements is connected in parallel with one of the second photosensitive elements, which is located in the same row and in the same side as the first photosensitive element and adjacent to the first photosensitive element.

[0036] Optionally, in some embodiments of the present application, the photosensitive device further includes a third photosensitive element;

[0037] where the third photosensitive element includes a third electrode, a third protection electrode, and a third photosensitive portion between the third electrode and the third protection electrode; the third electrode and the first electrode are located in the same layer and connected to each other, the third photosensitive portion and the first photosensitive portion are located in the same layer and arranged at intervals, and the third protection electrode is connected to either the source or the drain; and

[0038] the first photosensitive element, the second photosensitive element, and the third photosensitive element are respectively located in three of the intersection regions at three vertices of the same light emitting unit.

[0039] Another embodiment of the present application further involves a display panel, including: at least one photosensitive device, including a switching element, a first photosensitive element, and a second photosensitive element; where the display panel further includes:

[0040] a substrate;

[0041] a first conductive layer disposed on the substrate and including a first electrode and a second electrode connected to the first electrode;

[0042] a photosensitive layer disposed on one side of the first conductive layer away from the substrate and including a first photosensitive portion corresponding to the first electrode and a second photosensitive portion corresponding to the second electrode, where the first photosensitive portion and the second photosensitive portion are disposed at intervals, and the photosensitive layer further includes a metal oxide semiconductor;

[0043] an insulating layer disposed between the first conductive layer and the photosensitive layer;

[0044] a second conductive layer disposed on one side of the photosensitive layer away from the substrate and including a first protection electrode and a second protection electrode, where the

first protection electrode is electrically connected to the first photosensitive portion, and the second protection electrode is electrically connected to the second photosensitive portion; and [0045] a third conductive layer disposed on one side of the second conductive layer away from the substrate and including a source and a drain of the switching element, where the first protection electrode and the second protection electrode are both electrically connected to the source or the drain of the switching element;

[0046] where the first photosensitive element includes the first electrode, the first photosensitive portion, and the first protection electrode, and the second photosensitive element includes the second electrode, the second photosensitive portion, and the second protective electrode;

[0047] the display panel further includes:

[0048] a first active layer disposed between the substrate and the first conductive layer and including a first active portion of the switching element, where a source and a drain of the switching element are electrically connected to the first active portion;

[0049] where the display panel further includes: a plurality of light emitting units arranged in a plurality of rows in a first direction and arranged in a plurality of columns in a second direction;

[0050] where a plurality of intersecting regions are formed between the plurality of rows of light emitting units and the plurality of columns of light emitting units, and each of the intersecting regions is provided with at most one first photosensitive element or one second photosensitive elements.

[0051] Optionally, in some embodiments of the present application, the photosensitive device further includes: a storage capacitor including a first capacitor electrode and a second capacitor electrode at least partially overlapping the first capacitor electrode;

[0052] the first capacitor electrode is electrically connected to both the first electrode and the second electrode; and

[0053] the second capacitor electrode is electrically connected to both the first protection electrode and the second protection electrode, and at least partially overlapping the first capacitor electrode.

[0054] Optionally, in some embodiments of the present application, the display panel further includes: a fourth conductive layer including a conductive portion, where an orthographic projection of the conductive portion on the substrate at least partially overlaps an orthographic projection of at least one of the first electrode and the second electrode on the substrate.

[0055] Optionally, in some embodiments of the present application, the metal oxide semiconductor of the photosensitive layer has an electron mobility greater than or equal to $10 \text{ cm}^2/\text{Vs}$, and the insulating layer has a thickness of 5 nm to 15 nm.

[0056] Optionally, in some embodiments of the present application, the display panel further includes: a pixel driving circuit including both a first driving transistor and a second driving transistor;

[0057] the display panel further includes:

[0058] a second active layer disposed between the substrate and the photosensitive layer and including a second active portion of the first driving transistor, where the second active portion includes a polysilicon semiconductor;

[0059] a third active layer disposed between the second active layer and the photosensitive layer and including a third active portion of the second driving transistor, where the third active portion includes a metal oxide semiconductor; and

[0060] a conductive layer disposed between the second active layer and the third active layer and including a second gate of the first driving transistor, where the second gate at least partially overlaps the second active portion;

[0061] where the second conductive layer further includes a third gate of the second driving transistor, the third gate at least partially overlapping the third active portion.

[0062] Optionally, in some embodiments of the present application, the first active portion includes a polysilicon semiconductor;

[0063] the second active layer further includes the first active portion, and the conductive layer further includes a first gate of the switching element, the first gate at least partially overlapping the first active portion.

[0064] Optionally, in some embodiments of the present application, the first active portion includes a metal oxide semiconductor.

[0065] the third active layer further includes the first active portion, the conductive layer further includes a fourth gate of the switching element, and the second conductive layer further includes a first gate of the switching element, where the fourth gate at least partially overlaps the first active portion, and the first gate at least partially overlaps the first active portion.

[0066] Optionally, in some embodiments of the present application, the first photosensitive element and the second photosensitive element are respectively located in two adjacent ones of the intersecting regions located on the same side of the same light emitting unit.

Beneficial Effects

[0067] The present application provides a display panel. The display panel includes: at least one photosensitive device, where the photosensitive device includes the switching element, the first photosensitive element, and the second photosensitive element. The first photosensitive element includes the first electrode, the first photosensitive portion, and the first protection electrode, and the second photosensitive element includes the second electrode, the second photosensitive portion, and the second protective electrode. Since the first electrode is connected to the second electrode, and the first photosensitive portion and the second photosensitive portion are both connected to either the source or the drain of the switching element, the first photosensitive element and the second photosensitive element are designed in parallel to increase the photosensitive area of the photosensitive device, thereby improving the amount of the photoelectric signal and the detection sensitivity. Additionally, since the first electrode and the second electrode are disposed at the same layer and the first photosensitive portion and the second photosensitive portion are disposed at the same layer, the film layer structure of the photosensitive device is simplified. Further, since the insulating layer is provided between the first electrode and the first photosensitive portion, the barrier difference between the first electrode and the first photosensitive portion can be reduced, thereby allowing electron tunneling to form a conductive path. Similarly, since the insulating layer is provided between the second electrode and the second photosensitive portion, the barrier difference between the second electrode and the second photosensitive portion can be reduced, thereby allowing electron tunneling to form a conductive path. Therefore, the photosensitive performance of the photosensitive device can be improved.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0068] In order to more clearly illustrate the technical solutions in embodiments of the present application, the accompanying drawings depicted in the description of the embodiments will be briefly described below. It will be apparent that the accompanying drawings in the following description are merely some embodiments of the present application, and other drawings may be obtained from these drawings without creative effort by those skilled in the art.

[0069] FIG. 1 is a schematic diagram of a first structure of a display panel according to the present application.

[0070] FIG. 2 is a schematic circuit diagram of a photosensitive device according to the present application.

[0071] FIG. 3 is a schematic diagram of a second structure of a display panel according to the present application.

[0072] FIG. 4 is a schematic diagram of a third structure of a display panel according to the present

application.

[0073] FIG. 5 is a schematic diagram of a fourth structure of a display panel according to the present application.

[0074] FIG. 6 is a schematic diagram of a fifth structure of a display panel according to the present application.

[0075] FIG. 7 is a schematic diagram of a sixth structure of a display panel according to the present application.

[0076] FIG. 8 is a schematic diagram of a seventh structure of a display panel according to the present application.

[0077] FIG. 9 is a schematic diagram of an eighth structure of a display panel according to the present application.

[0078] FIG. 10 is a schematic diagram of a ninth structure of a display panel according to the present application.

[0079] FIG. 11 is a first plan view of a display panel according to the present application.

[0080] FIG. 12 is a second plan view of a display panel according to the present application.

[0081] FIG. 13 is a third plan view of a display panel according to the present application.

DETAILED DESCRIPTION

[0082] Technical solutions in embodiments of the present application will be clearly and completely described below in conjunction with drawings in the embodiments of the present application. Obviously, the described embodiments are only a part of embodiments of the present application, rather than all the embodiments. Based on the embodiments in the present application, all other embodiments obtained by those skilled in the art without creative work fall within the protection scope of the present application.

[0083] In the description of the present application, it should be understood that the terms “first” and “second” are for illustrative purposes only and are not to be construed as indicating or imposing a relative importance or implicitly indicating the number of technical features indicated. Thus, a feature that limited by “first” and “second” may expressly or implicitly include at least one of the features, and therefore cannot be construed as a limitation on the present application. Additionally, it should be noted that the terms “connected to” and “connection” should be understood in a broad sense, unless otherwise clearly specified and defined. For example, it can be a mechanical connection, or an electrical connection; it can be directly connected or indirectly connected through an intermediary, it can also be the connection between two elements. Those ordinary skilled in the art can understand the specific meanings of the above terms in the present application according to specific situations.

[0084] The present application provides a display panel, which is described in detail below. It should be noted that the description order of the following embodiments of the present application is not intended to limit the preferred order of the embodiments.

[0085] Please refer to FIG. 1, which is a schematic diagram of a first structure of a display panel according to the present application. In some embodiments of the present application, the display panel 100 includes at least one photosensitive device ST. The photosensitive device ST includes a switching element T, a first photosensitive element S1, and a second photosensitive element S2.

[0086] The display panel 100 further includes a substrate 10, a first conductive layer 25, an insulating layer 26, a photosensitive layer 27, a second conductive layer 28, and a third conductive layer 30.

[0087] The first conductive layer 25 is disposed on the substrate 10. The first conductive layer 25 includes a first electrode 251 and a second electrode 252 connected to the first electrode 251.

[0088] A photosensitive layer 27 is disposed on one side of the first conductive layer 25 away from the substrate 10. The photosensitive layer 27 includes a first photosensitive portion 271 corresponding to the first electrode 251 and a second photosensitive portion 272 corresponding to the second electrode 252. The photosensitive layer 27 includes a metal oxide semiconductor. The

first photosensitive portion **271** and the second photosensitive portion **272** are provided at intervals. [0089] An insulating layer **26** is disposed between the first conductive layer **25** and the photosensitive layer **27**. For example, the insulating layer **26** includes a first insulating portion **261** and a second insulating portion **262**. The first insulating portion **261** is located at least between the first electrode **251** and the first photosensitive portion **271**. The second insulating portion **262** is located at least between the second electrode **252** and the second photosensitive portion **272**.

[0090] A second conductive layer **28** is disposed on one side of the photosensitive layer **27** away from the substrate **10**. The second conductive layer **28** includes a first protective electrode **281** and a second protective electrode **282**. The first protection electrode **281** is electrically connected to the first photosensitive portion **271**. The second protection electrode **282** is electrically connected to the second photosensitive portion **272**.

[0091] A third conductive layer **30** is disposed on one side of the second conductive layer **28** away from the substrate **10**. The third conductive layer **30** includes a source **301** and a drain **302** of the switching element T. The first protection electrode **281** and the second protection electrode **282** are both connected to either the source **301** or the drain **302**. For example, the first protection electrode **281** and the second protection electrode **282** are both connected to the source **301**. For another example, both the first protection electrode **281** and the second protection electrode **282** are both connected to the drain **302**.

[0092] The first photosensitive element S1 includes the first electrode **251**, the first insulating portion **261**, the first photosensitive portion **271**, and the first protective electrode **281**. The second photosensitive element S2 includes the second electrode **252**, the second insulating portion **262**, the second photosensitive portion **272**, and the second protective electrode **282**.

[0093] The first electrode **251** may be directly connected to the second electrode **252**. That is, the first electrode **251** and the second electrode **252** are designed as an integrated structure. The first electrode **251** may also be connected to the second electrode **252** by either a bridge line or a via hole, which is not specifically limited in the present application. In each of the following embodiments of the present application, that the first electrode **251** is directly connected to the second electrode **252** is taken as an example for illustration, but cannot be understood as a limitation of the present application.

[0094] In the embodiments of the present application, the photosensitive device ST includes at least two photosensitive elements, i.e., a first photosensitive element S1 and a second photosensitive element S2, respectively. Since the first electrode **251** is connected to the second electrode **252**, and the first protection electrode **281** and the second protection electrode **282** are both connected to either the source **301** or the drain **302**, so that parallel connection between the first photosensitive element S1 and the second photosensitive element S2 is implemented to increase the photosensitive area of the photosensitive device ST, thereby improving the amount of the photoelectric signal and the detection sensitivity. Additionally, since the first electrode **251** and the second electrode **252** are disposed at the same layer, the first photosensitive portion **271** and the second photosensitive portion **272** are disposed at the same layer, and the first protection electrode **281** and the second protection electrode **282** are disposed at the same layer, the film layer structure of the photosensitive device ST is simplified. Further, since the first insulating portion **261** is provided between the first electrode **251** and the first photosensitive portion **271**, the barrier difference between the first electrode **251** and the first photosensitive portion **271** can be reduced, thereby allowing electron tunneling to form a conductive path. Similarly, since the second insulating portion **262** is provided between the second electrode **252** and the second photosensitive portion **272**, the barrier difference between the second electrode **252** and the second photosensitive portion **272** can be reduced, thereby allowing electron tunneling to form a conductive path. Thus, compared with a case where a high work function metal is required to form a metal-semiconductor junction in the related art, the photosensitive device ST can be formed in the present application by using the first conductive layer **25** of a conventional material, thereby improving the photosensitive

performance of the photosensitive device ST.

[0095] It may be understood that, when a wiring space in the display panel **100** is limited, the photosensitive area of each of the first photosensitive element **S1** and the second photosensitive element **S2** can be relatively small in design, the light signal absorbed by each of the first photosensitive portion **271** and the second photosensitive portion **272** is relatively small, and the current signal generated thereby is relatively small. In some embodiments of the present application, one photosensitive device ST includes at least the first photosensitive element **S1** and the second photosensitive element **S2**, which are designed in parallel, so that an overall photosensitive area of the photosensitive device ST is increased, the amount of the photoelectric signal is increased, and the detection sensitivity is improved.

[0096] It should be noted that, in the embodiments of the present application, that the photosensitive device ST includes at least two photosensitive elements (a first photosensitive element **S1** and a second photosensitive element **S2**) connected in parallel is taken as an example for illustration, but cannot be understood as a limitation of the present application. For example, one photosensitive device ST may include three photosensitive elements, five photosensitive elements, or more in parallel.

[0097] In the embodiments of the present application, the switching element T further includes a first gate **253** and a first active portion **231**. The first active portion **231** is provided corresponding to the first gate **253**. The source **301** and the drain **302** are connected to the first active portion **231**, respectively. The film structure of the first gate **253**, the first active portion **231**, the source **301**, and the drain **302** will be illustrated in the following embodiments, of which details are not repeatedly described herein.

[0098] In the embodiments of the present application, the photosensitive device ST further includes at least one storage capacitor C. A first capacitor plate of the storage capacitor C is electrically connected to both the first electrode **251** and the second electrode **252**. A second capacitor plate of the storage capacitor C is connected to either the source **301** or the drain **302**.

[0099] For example, the photosensitive device ST includes only one storage capacitor C, and the first photosensitive element **S1** and the second photosensitive element **S2** share one storage element C, thereby further simplifying the structure of the photosensitive device ST.

[0100] Specifically, please refer to FIGS. 1-2, FIG. 2 is a schematic circuit diagram of a photosensitive device according to the present application. The photosensitive device ST includes a first photosensitive element **S1**, a second photosensitive element **S2**, a storage capacitor C, and a switching element T.

[0101] The first electrode **251** and the second electrode **252** are connected to a bias voltage VBias. When light enters the first photosensitive element **S1** and the second photosensitive element **S2** in the photosensitive device ST, the first photosensitive portion **271** in the first photosensitive element **S1** and the second photosensitive portion **272** in the second photosensitive element **S2** absorb light signals and convert the received light signals into electrical signals. The electric signals are stored in the storage capacitor C, and after the storage capacitor C is filled, the switching element T is turned on and the storage capacitor C is discharged, so that the signals generated by the first photosensitive element **S1** and the second photosensitive element **S2** can be transmitted to a detection signal line (not shown), via which the signals are transmitted to the corresponding circuits for processing, thereby realizing the detection of the light intensity.

[0102] In some embodiments of the present application, materials of the first active portion **231** may be a monocrystalline silicon, a low-temperature polycrystalline silicon, or a metal oxide semiconductor. The metal oxide semiconductor may be Indium Gallium Zinc Oxide (IGZO), Indium Gallium Zinc Tin Oxide (IGZTO), Indium Zinc Oxide (IZO), Indium Gallium Oxide (IGO), Indium Gallium Tin Oxide (IGTO), Indium Zinc Tin Oxide (IZTO), Indium Tin Oxide (ITO), Zinc Aluminum Tin Oxide (ATZO), Zinc Aluminum Indium Oxide (AIZO), or the like.

[0103] In some embodiments, the metal oxide semiconductor of the photosensitive layer **27** has an

electron mobility greater than or equal to 10 cm.sup.2/Vs. Specifically, the materials of the first photosensitive portion **271** and the second photosensitive portion **272** are the IGZO. The IGZO has a high mobility. The IGZO has a good photosensitivity feature and a low resistance in the visible light band, so that the photosensitivity of the photosensitive device ST can be improved.

[0104] In some embodiments, the materials of the first active portion **231**, the first photosensitive portion **271**, and the second photosensitive portion **272** are all IGZO. Thus, the first photosensitive portion **271** and the second photosensitive portion **272** may be shared with an IGZO substrate film layer.

[0105] In some embodiments, the materials of the first active portion **231**, the first photosensitive portion **271**, and the second photosensitive portion **272** are different from each other. For example, the materials of the first photosensitive portion **271** and the second photosensitive portion **272** are indium zinc gallium oxide, and the material of the first active portion **231** may be monocrystalline silicon, low-temperature polycrystalline silicon, or other oxide semiconductor material other than IGZO. Thus, performance requirements of the switching element T, the first photosensitive element S1, and the second photosensitive element S2 can be simultaneously met.

[0106] In some embodiments, an orthographic projection of the first protection electrode **281** on the substrate **10** overlaps an orthographic projection of the first photosensitive portion **271** on the substrate **10**, and an orthographic projection of the second protection electrode **282** on the substrate **10** overlaps an orthographic projection of the second photosensitive portion **272** on the substrate **10**. Thus, the same photomask can be used for patterning to form the photosensitive layer **27** and the second conductive layer **28**, thereby simplifying the process. Of course, the present application is not limited thereto.

[0107] In the embodiments of the present application, the second photosensitive element S2 is located on one side of the first photosensitive element S1 away from the drain **302**. The first conductive layer **25** further includes a first capacitor electrode of the storage capacitor C. The first capacitor electrode is electrically connected to both the first electrode **251** and the second electrode **252**. The third conductive layer **30** further includes a second capacitor electrode of the storage capacitor C. The second capacitor electrode is electrically connected to both the first protection electrode **281** and the second protection electrode **282**. The second capacitor electrode at least partially overlaps the first capacitor electrode.

[0108] Specifically, the first capacitor electrode is a portion of the drain **302**, and the second capacitor electrode is a portion of the first electrode **251**. An orthographic projection of the drain **302** on the substrate **10** at least partially overlaps an orthographic projection of the first electrode **251** on the substrate **10**. That is, the drain **302** is used with at least the first electrode **251** to constitute the storage capacitor C. Since the first electrode **251** is connected to the second electrode **252**, the first photosensitive element S1 and the second photosensitive element S2 share the storage capacitor C.

[0109] In the embodiments of the present application, the display panel **100** further includes a first active layer **23** and a gate insulating layer **24**. The first active layer **23** is disposed on one side of the first conductive layer **25** close to the substrate **10**. The gate insulating layer **24** is disposed between the first active layer **23** and the first conductive layer **25**. The first active layer **23** includes a first active portion **231**. The first conductive layer **25** further includes a first gate **253**. The first active portion **231** is provided corresponding to the first gate **253**.

[0110] According to the embodiments of the present application, the first gate **253** is disposed at the same layer as the first electrode **251** and the second electrode **252**, so that one photomask can be saved, and the process can be simplified. At the same time, the thickness of the display panel **100** is reduced.

[0111] In one embodiment, the display panel **100** further includes an interlayer insulating layer **29**. The interlayer insulating layer **29** is located between the first conductive layer **25** and the third conductive layer **30**. The interlayer insulating layer **29** has a first via hole **29a**, a second via hole

29b, and a third via hole **29c**. The first via hole **29a** extends through the interlayer insulating layer **29** and to one side of the first active portion **231** away from the substrate **10**. The source **301** is connected to the first active portion **231** through the first via hole **29a**. The second via hole **29b** extends through the interlayer insulating layer **29** and to one side of the first active portion **231** away from the substrate **10**. The drain **302** is connected to the first active portion **231** through the second via hole **29b**. The third via hole **29c** exposes one side surface of the first protection electrode **281** away from the substrate **10**. The drain **302** is connected to the first protection electrode **281** through the third via hole **29c**.

[0112] The first active portion **231** includes a source region, a drain region, and a channel region (not shown) between the source region and the drain region. The source **301** is connected to the source region. The drain **302** is connected to the drain region. Since the source region and the drain region are conductive by ion doping or the like, the conductivity of the source **301** and the drain **302** as well as the first active portion **231** can be improved.

[0113] It should be noted that, in the embodiments of the present application, that the switching element T is used as a top-gate transistor is taken as an example for illustration, but cannot be understood as a limitation of the present application. In some embodiments of the present application, the switching element T may also be a bottom-gate transistor or a double gate transistor.

[0114] In the embodiments of the present application, the substrate **10** may include, but not limited to, a base substrate **11**, a barrier layer **12**, a first insulating layer **13**, and a second insulating layer **14**, which are sequentially stacked from bottom to top. The material of the base substrate **11** may be glass or a flexible material. The materials of the barrier layer **12**, the first insulating layer **13**, and the second insulating layer **14** may be silicon oxide, silicon nitride, or the like. The barrier layer **12**, the first insulating layer **13**, and the second insulating layer **14** may serve as water and oxygen barriers.

[0115] In the embodiments of the present application, the first conductive layer **25** is made of a material having high conductivity and a good light shielding property. For example, the material of the first conductive layer **25** may be molybdenum, titanium, molybdenum/copper (stack), molybdenum/titanium (stack), titanium/aluminum (stack), or the like. According to the embodiments of the present application, the first electrode **251** and the second electrode **252** are made of a conductive material having a light shielding property, so that the light incident from the substrate **10** onto the first photosensitive element **S1** and the second photosensitive element **S2** can be prevented, thereby improving the detection accuracy of the photosensitive device **ST**.

[0116] In the embodiments of the present application, the material of the insulating layer **26** may be silicon nitride, silicon oxide, or the like. The thickness of the insulating layer **26** is very thin, typically around 10 nm, such as 5 nm to 15 nm. When the first photosensitive element **S1** and the second photosensitive element **S2** are operated, electrons may pass through the first photosensitive element **S1** and the second photosensitive element **S2**.

[0117] In the embodiments of the present application, the materials of the gate insulating layer **24** and the interlayer insulating layer **29** may be silicon oxide, silicon nitride, trialuminum oxide, and a stack thereof.

[0118] In the embodiments of the present application, the second conductive layer **28** is a transparent conductive material, so as to ensure that light can be incident onto the first photosensitive element **S1** and the second photosensitive element **S2** and improve the detection sensitivity of the photosensitive device **ST**. For example, the material of the second conductive layer **28** may be ITO, IZO, or the like.

[0119] In the embodiments of the present application, the display panel **100** further includes a fourth conductive layer **21** and a buffer layer **22**. The fourth conductive layer **21** is disposed on one side of the first active layer **23** close to the substrate **10**. The buffer layer **22** is disposed between the first active layer **23** and the fourth conductive layer **21**. The fourth conductive layer **21** includes

a light shielding portion **211**. The light shielding portion **211** is provided corresponding to the first active unit **231**. For example, an orthographic projection of the light shielding portion **211** on the substrate **10** at least covers an orthographic projection of the channel portion of the first active portion **231** on the substrate **10**.

[0120] The light shielding portion **211** may block light incident from the substrate **10** in a direction away from the light shielding portion **211**, thereby reducing interference of external light to the first active portion **231**, and further improving the operation performance of the photosensitive device ST.

[0121] In some embodiments, the light shielding portion **211** may be further connected to either the source **301** or the drain **302** to form an equipotential, so that changes in the voltage on the light shielding portion **211** can be prevented from affecting the electrical performance of the first active portion **231**.

[0122] In the embodiments of the present application, the third conductive layer **30** may further include an input electrode **303**. The input electrode **303** is connected to either the first electrode **251** or the second electrode **252**. The input electrode **303** is used to be connected to a bias voltage Vbias.

[0123] For example, the interlayer insulating layer **29** further includes a connection hole **29d**. The connection hole **29d** extends to one side surface of the second electrode **252** away from the substrate **10**. The input electrode **303** is connected to the second electrode **252** through the connection hole **29d**.

[0124] In the embodiments of the present application, the first conductive layer **25** further includes at least one scan line, which may be multiplexed in a time-division manner as the first electrode **251** and the second electrode **252**. Time division multiplexing means that the scan line may be used to transmit a scan signal, or may be used as the first electrode **251** and the second electrode **252** to transmit a bias voltage Vbias. Specifically, the scan line is connected to a gate driving circuit and the signal line for supplying the bias voltage Vbias, respectively. When fingerprint recognition is performed, the signal line transmits the bias voltage Vbias to the scan line. When the display is performed, the gate driving circuit supplies the scan signal to the scan line.

[0125] Thus, the wiring in the display panel **100** can be reduced, and the sizes of the first photosensitive element S1 and the second photosensitive element S2 can be increased, thereby increasing the photosensitive area. It should be noted that, when a plurality of photosensitive devices ST are provided in the display panel **100**, each scan line is multiplexed as only the first electrode **251** and the second electrode **252** connected to the first electrode **251** in one of the photosensitive devices ST.

[0126] Please refer to FIG. 3, which is a schematic diagram of a second structure of a display panel according to the present application. A difference from the display panel **100** shown in FIG. 1 lies in that, in the embodiments of the present application, the display panel **100** includes a fourth conductive layer **21**. The fourth conductive layer **21** includes a conductive portion **212**. An orthographic projection of the conductive portion **212** on the substrate **10** at least partially overlaps an orthographic projection of the drain **302** on the substrate **10**. That is, the conductive portion **212** is used along with the drain **302** to constitute the storage capacitor C.

[0127] In the embodiments of the present application, the conductive portion **212** is used along with the drain **302** to constitute the storage capacitor C, so that the length of the first electrode **251** can be reduced in the direction from the switching element T to the first photosensitive element S1. In such a case that the cross sectional area of the first electrode **251** is constant, the length of the first electrode **251** is reduced, thereby reducing the resistance of the first electrode **251** and reducing the load of the first photosensitive element S1 and the second photosensitive element S2.

[0128] Additionally, the fourth conductive layer **21** may further include a light shielding portion **211**. The conductive portion **212** and the light shielding portion **211** are provided at the same layer, so that one photomask can be saved, and the process can be simplified. Of course, in some

embodiments, the conductive portion **212** can be also provided at a different layer than the light shielding portion **211**.

[0129] Please refer to FIG. **4**, which is a schematic diagram of a third structure of a display panel according to the present application. A difference from the display panel **100** shown in FIG. **1** lies in that, in the embodiments of the present application, the fourth conductive layer **21** further includes a conductive portion **212**. An orthographic projection of the conductive portion **212** on the substrate **10** at least partially overlaps an orthographic projection of the first electrode **251** and/or the second electrode **252** on the substrate **10**. That is, the conductive portion **212** is used along with the first electrode **251** and/or the second electrode **252** to constitute the storage capacitor C.

[0130] It should be understood in the embodiments of the present application that the conductive portion **212** is used along with the first electrode **251** and/or the second electrode **252** to constitute the storage capacitor C, so that the extension length of the drain **302** from the switching element T to the first photosensitive element S1 can be reduced, thereby reducing the distance between the switching element T and the first photosensitive element S1 and reducing the wiring in the display panel **100**. Since the conductive portion **212** is located below the first electrode **251** and/or the second electrode **252**, the capacitance value of the storage capacitor C can be adjusted by adjusting the area of the conductive portion **212**, so that no additional wiring space is occupied. As a result, the wiring space of the first photosensitive element S1 and the second photosensitive element S2 can be increased, the photosensitive area of the photosensitive device is further increased, and the sensitivity is improved.

[0131] Please refer to FIG. **5**, which is a schematic diagram of a fourth structure of a display panel according to the present application. A difference from the display panel **100** shown in FIG. **1** lies in that, in the embodiments of the present application, the first active layer **23** includes a first active portion **231** and a conductive electrode portion **232**.

[0132] The first active portion **231** is provided corresponding to the first gate **253**. The source **301** and the drain **302** are connected to the first active portion **231**, respectively. An orthographic projection of the electrode portion **232** on the substrate **10** partially overlaps an orthographic projection of the first electrode **251** and/or the second electrode **252** on the substrate **10**.

[0133] The electrode portion **232** may be subjected to a conductive processing by a process such as ion doping to improve the conductivity of the electrode portion **232**.

[0134] In the embodiments of the present application, the conductive electrode portion **232** is used along with the first electrode **251** and/or the second electrode **252** to constitute the storage capacitor C. On one hand, it is possible to reduce the distance between the switching element T and the first photosensitive element S1, reduce the wiring in the display panel **100**, and increase the wiring space of the first photosensitive element S1 and the second photosensitive element S2, thereby further improving the photosensitive area of the photosensitive device, and improving the sensitivity. On the other hand, since only the buffer layer **22** is provided between the electrode portion **232** and the first electrode **251** (the second electrode **252**), the distance between two electrode plates of the storage capacitor C is reduced, thereby increasing the capacitance value of the storage capacitor C.

[0135] Please refer to FIG. **6**, which is a schematic diagram of a fifth structure of a display panel according to the present application. A difference from the display panel **100** shown in FIG. **1** lies in that, in the embodiments of the present application, the insulating layer **26** further includes a third insulating portion **263**. An orthographic projection of the first insulating portion **261** and the second insulating portion **262** on the substrate **10** overlaps an orthographic projection of the first electrode **251** and the second electrode **252** on the substrate **10**. An orthographic projection of the third insulating portion **263** on the substrate **10** overlaps an orthographic projection of the first gate **253** on the substrate **10**.

[0136] Patterning of the first conductive layer **25** and the insulating layer **26** can be realized in the embodiments of the present application by using the same photomask, thereby simplifying the

process. Further, since the first insulating portion **261** covers the first electrode **251** and the second electrode **252**, and the third insulating portion **263** covers the first gate **253**, the insulating layer **26** can protect the first electrode **251**, the second electrode **252**, and the first gate **253** when the photosensitive layer **27** is formed, thereby improving the stability of the photosensitive device. [0137] Please refer to FIG. 7, which is a schematic diagram of a sixth structure of a display panel according to the present application. A difference from the display panel **100** shown in FIG. 1 lies in that, in the embodiments of the present application, the display panel **100** further includes a pixel driving circuit. The pixel driving circuit further includes a first driving transistor **T1** and a second driving transistor **T2**.

[0138] The display panel **100** further includes a second active layer **32**, a third active layer **33**, and a conductive layer **34**.

[0139] The second active layer **32** is disposed between the substrate **10** and the photosensitive layer **27**. The second active layer **32** includes a second active portion **321** of the first driving transistor **T1**. The second active portion **321** includes a polysilicon semiconductor.

[0140] The third active layer **33** is disposed between the second active layer **32** and the photosensitive layer **27**. The third active layer **33** includes a third active portion **331** of the second driving transistor **T2**. The third active portion **331** includes a metal oxide semiconductor.

[0141] The conductive layer **34** is disposed between the second active layer **32** and the third active layer **33**. The conductive layer **34** includes a second gate **341** of the first driving transistor **T1**. The second gate **341** at least partially overlaps the second active portion **321**.

[0142] The second conductive layer **28** further includes a third gate **283** of the second driving transistor **T2**. The third gate **283** at least partially overlaps the third active portion **331**.

[0143] In some embodiments of the present application, still referring to FIG. 7, the first active portion **231** includes a polysilicon semiconductor. The second active layer **32** further includes a first active portion **231**. The conductive layer **34** further includes a first gate **253** of the switching element **T**. The first gate **253** at least partially overlaps the first active portion **231**. That is, the first active layer **23** and the second active layer **32** are the same active layer.

[0144] The first conductive layer **25** further includes a third electrode **255**. The third electrode **255** and the second gate **341** are used along with the first insulating layer **13** between the third electrode **255** and the second gate **341** to constitute a capacitor. The conductive layer **34** further includes a fifth gate **343** of the second driving transistor **T2**, that is, the second driving transistor **T2** is of a double gate structure.

[0145] The third conductive layer **30** further includes a first source **304** and a first drain **305** of the first driving transistor **T1**, and a second source **306** and a second drain **307** of the second driving transistor **T2**. The first drain **305** may be connected to the second source **306** by a via hole.

[0146] In some embodiments, please refer to FIG. 8, which is a schematic diagram of a seventh structure of a display panel according to the present application. A difference from the display panel **100** shown in FIG. 7 lies in that, in the embodiments of the present application, the first active layer **231** includes a metal oxide semiconductor. The third active layer **33** further includes a first active portion **231**.

[0147] The conductive layer **34** further includes a fourth gate **342** of the switching element **T**. The second conductive layer **28** further includes a first gate **253** of the switching element **T**. The fourth gate **342** at least partially overlaps the first active portion **231**. The first gate **253** at least partially overlaps the first active portion **231**. That is, the switching element **T** is of a double gate structure.

[0148] Please refer to FIG. 9, which is a schematic diagram of an eighth structure of a display panel according to the present application. A difference from the display panel **100** shown in FIG. 1 lies in that, in the embodiments of the present application, the photosensitive layer **27** further includes a first active portion **231** of the switching element **T**. The gate insulating layer **24** has an opening **24a**. The opening **24a** exposes one side surface of the first photosensitive portion **271** and the second photosensitive portion **272** away from the substrate **10**. The first protection electrode **281** and the

second protection electrode **282** are provided within the opening **24a**.

[0149] In some embodiments of the present application, the first active portion **231**, the first photosensitive portion **271**, and the second photosensitive portion **272** are provided at the same layer, which can simplify the manufacturing process. At the same time, the gate insulating layer **24** is formed, and then the first protection electrode **281** and the second protection electrode **282** are formed, so that damage to the first active portion **231** can be avoided when the second conductive layer **28** is patterned, thereby improving the stability of the switching element T.

[0150] Further, in some embodiments, still referring to FIG. **9**, the display panel **100** further includes a pixel circuit. The pixel circuit includes a first driving transistor T1 and a second driving transistor T2.

[0151] The photosensitive layer **27** further includes a first active portion **231** of the switching element T and a third active portion **331** of the second driving transistor T2. The first active portion **231** of the switching element T, the third active portion **331** of the second driving transistor T2, the first photosensitive portion **271**, and the second photosensitive portion **272** are provided in the same layer.

[0152] Specifically, the display panel **100** further includes a second active layer **32**, a conductive layer **34**, and a fifth conductive layer **31**. The second active layer **32** is disposed between the substrate **10** and the photosensitive layer **27**. The conductive layer **34** is disposed between the second active layer **32** and the photosensitive layer **27**. The fifth conductive layer **31** is disposed on one side of the gate insulating layer **24** away from the substrate **10**. The second active layer **32** includes a second active portion **321** of the first driving transistor T1. The second active portion **321** includes a polysilicon semiconductor. The conductive layer **34** includes a second gate **341** of the first driving transistor T1 and an electrode portion **232**. The second gate **341** at least partially overlaps the second active portion **321**. The fifth conductive layer **31** further includes a first gate **253** of the switching element T and a third gate **283** of the second driving transistor T2.

[0153] The first conductive layer **25** further includes a fourth gate **342** of the switching element T, a fifth gate **343** of the second driving transistor T2, and a third electrode **255**. The third conductive layer **30** further includes a first source **304** and a first drain **305** of the first driving transistor T1, and a second source **306** and a second drain **307** of the second driving transistor T2. The first drain **305** may be connected to the second source **306** by a via hole.

[0154] Please refer to FIG. **10**, which is a schematic diagram of a ninth structure of a display panel according to the present application. A difference from the display panel **100** shown in FIG. **9** lies in that, in the embodiments of the present application, the second active portion **321** of the first driving transistor T1 and the first active portion **231** of the switching element T are provided in the same layer. The third active portion **331** of the second driving transistor T2, the first photosensitive portion **271**, and the second photosensitive portion **272** are provided in the same layer.

[0155] Specifically, the second active layer **32** includes a second active portion **321** of the first driving transistor T1 and a first active portion **231** of the switching element T. The conductive layer **34** includes a second gate **341** of the first driving transistor T1 and a first gate **253** of the switching element T.

[0156] The embodiments of the present application employ Low Temperature Poly-Oxide (LTPO) technology, which can reduce power consumption of the display panel **100**. Meanwhile, the switching element T and the photosensitive device ST are manufactured in the same process as the in-plane LTPO structure, so that the thickness of the display panel **100** can be reduced, and the manufacturing process can be simplified.

[0157] Please refer to FIG. **11**, which is a first plan view of a display panel according to the present application. The display panel **100** further includes a plurality of light emitting units **40**. The plurality of light emitting units **40** are arranged in a plurality of rows in a first direction Y, and are arranged in a plurality of columns in a second direction X. The first direction Y intersects the second direction X. For example, the first direction Y is perpendicular to the second direction X.

[0158] A plurality of intersecting regions **40a** are formed between the plurality of rows of light emitting units **40** and the plurality of columns of light emitting units **40**. Each of the intersecting regions **40a** is provided with at most one first photosensitive element **S1** or one second photosensitive elements **S2**.

[0159] In the embodiments of the present application, the first photosensitive element **S1** and the second photosensitive element **S2** are disposed in different intersecting regions **40a** so that the first photosensitive element **S1** and the second photosensitive element **S2** are both staggered from the light emitting unit **40** to avoid affecting the display effect of the display panel. Further, by disposing the first photosensitive element **S1** and the second photosensitive element **S2** in different intersecting regions **40a**, the wiring space of the first photosensitive element **S1** and the second photosensitive element **S2** can be increased, and the photosensitive area of the first photosensitive element **S1** and the second photosensitive element **S2** can be increased, thereby further increasing the amount of the electric signal.

[0160] In the embodiment of the present application, each of the intersection regions **40a** may be provided with the first photosensitive element **S1** or the second photosensitive element **S2** to improve test sensitivity. Of course, the distribution densities of the first photosensitive element **S1** and the second photosensitive element **S2** may be set according to actual product requirements.

[0161] As shown in FIG. **11**, the first photosensitive element **S1** and the second photosensitive element **S2** are located in two adjacent ones of the intersecting regions **40a**, respectively. The two adjacent intersecting regions **40a** are located on the same side of the same light emitting unit **40**.

[0162] As a result, the distance between the first photosensitive element **S1** and the second photosensitive element **S2** in the same photosensitive device **ST** can be reduced, thereby facilitating parallel connection between the first photosensitive element **S1** and the second photosensitive element **S2** and reducing the wiring length.

[0163] In some embodiments of the present application, the display panel **100** includes a plurality of photosensitive devices **ST**. A plurality of first photosensitive elements **S1** and a plurality of second photosensitive elements **S2** in the same row are alternately arranged. A plurality of first photosensitive elements **S1** and a plurality of second photosensitive elements **S2** in the same column are alternately arranged. Each of the first photosensitive elements **S1** is connected in parallel with one of the second photosensitive elements **S2**, which is located in the same row and in the same side as the first photosensitive element and adjacent to the first photosensitive element. Alternatively, each of the first photosensitive elements **S1** is connected in parallel with one of the second photosensitive elements **S2**, which is located in the same column and in the same side as the first photosensitive element and adjacent to the first photosensitive element.

[0164] For example, as shown in FIG. **11**, each of the first photosensitive elements **S1** is connected in parallel with one of the second photosensitive elements **S2**, which is located in the same row as the first photosensitive element **S1** and located at the right of the first photosensitive element **S1** and adjacent to the first photosensitive element **S1**. Of course, each of the first photosensitive elements **S1** may be also connected in parallel with one of the second photosensitive elements **S2**, which is located in the same row as the first photosensitive element **S1** and located at the left of the first photosensitive element **S1** and adjacent to the first photosensitive element **S1**.

[0165] It should be understood that the embodiments of the present application enable the plurality of photosensitive devices **ST** to be staggered in the display panel **100**, the distribution of the plurality of photosensitive devices **ST** in the display panel **100** is more uniform, and the uniformity of the detection sensitivity of the display panel **100** can be improved.

[0166] In some embodiments of the present application, as shown in FIG. **12**, a plurality of first photosensitive elements **S1** and a plurality of second photosensitive elements **S2** located in the same row are alternately arranged, and the first photosensitive elements **S1** or the second photosensitive elements **S2** are located in the same column. Each of the first photosensitive elements **S1** is connected in parallel with one of the second photosensitive elements **S2**, which is

located in the same row and in the same side as the first photosensitive element and adjacent to the first photosensitive element.

[0167] It should be understood that the embodiments of the present application make the positional relationship of the first photosensitive element **S1** and the second photosensitive element **S2** in each of the photosensitive devices **ST** identical, thereby improving the structural regularity of the plurality of photosensitive devices **S** and reducing the process difficulty.

[0168] In some embodiments of the present application, as shown in FIGS. **1** and **13**, the photosensitive device **ST** further includes a third photosensitive element **S3** (not shown in FIG. **1**).

[0169] The third photosensitive element **S3** includes a third electrode, a third protection electrode, and a third photosensitive portion between the third electrode and the third protection electrode. The third electrode, the first electrode **251** and the second electrode **252** are located at the same layer and connected to each other. The third photosensitive portion, the first photosensitive portion **271**, and the second photosensitive portion **272** are located at the same layer and disposed at intervals. The third protection electrode, the first protection electrode **281**, and the second protection electrode **282** are located at the same layer. The third protection electrode is connected to the source **301** or the drain **302**.

[0170] The first photosensitive element **S1**, the second photosensitive element **S2**, and the third photosensitive element **S3** are respectively located in three ones of the intersection regions **40a** at three vertices of the same light emitting unit.

[0171] Specifically, in the same column of the light emitting units **40**, for any two adjacent ones of the photosensitive devices **ST**, two first photosensitive elements **S1** are arranged diagonally, two second photosensitive elements **S2** are arranged diagonally, and two third photosensitive elements **S3** are located in the same row, thereby improving the distribution uniformity of the plurality of photosensitive devices **ST**.

[0172] The embodiments of the present application provide that the photosensitive device **ST** includes a first photosensitive element **S1**, a second photosensitive element **S2**, and a third photosensitive element **S3**, thereby further improving the photosensitive area of the photosensitive device **ST**. By arranging the first photosensitive element **S1**, the second photosensitive element **S2**, and the third photosensitive element **S3** in three ones of the intersecting regions **40a** at three vertices of the same light emitting unit **40**, respectively, it is possible to reduce the distance between any two ones of the first photosensitive element **S1**, the second photosensitive element **S2**, and the third photosensitive element **S3** in the same photosensitive device **ST**, thereby facilitating parallel connection between any two ones of the first photosensitive element **S1**, the second photosensitive element **S2**, and the third photosensitive element **S3**, and reducing the wiring length.

[0173] The display panels provided in the embodiments of the present application are described in detail above. A specific example is used herein to describe a principle and an implementation of the present application. The description of the foregoing embodiments is merely used to help understand a method and a core idea of the present application. In addition, an ordinary person skilled in the art may make changes in a specific implementation manner and an application scope according to an idea of the present application. In conclusion, content of this specification should not be construed as a limitation on the present application.

Claims

1. A display panel, comprising: at least one photosensitive device, the at least one photosensitive device including a switching element, a first photosensitive element, and a second photosensitive element; wherein the display panel further comprises: a substrate; a first conductive layer disposed on the substrate and including a first electrode and a second electrode connected to the first electrode; a photosensitive layer disposed on one side of the first conductive layer away from the substrate and including a first photosensitive portion corresponding to the first electrode and a

second photosensitive portion corresponding to the second electrode, wherein the first photosensitive portion and the second photosensitive portion are disposed at intervals, and the photosensitive layer further includes a metal oxide semiconductor; an insulating layer disposed between the first conductive layer and the photosensitive layer; a second conductive layer disposed on one side of the photosensitive layer away from the substrate and including a first protection electrode and a second protection electrode, wherein the first protection electrode is electrically connected to the first photosensitive portion, and the second protection electrode is electrically connected to the second photosensitive portion; and a third conductive layer disposed on one side of the second conductive layer away from the substrate and including a source and a drain of the switching element, wherein the first protection electrode and the second protection electrode are both electrically connected to the source or the drain of the switching element; wherein the first photosensitive element includes the first electrode, the first photosensitive portion, and the first protection electrode, and the second photosensitive element includes the second electrode, the second photosensitive portion, and the second protective electrode.

2. The display panel of claim 1, further comprising: a first active layer disposed between the substrate and the first conductive layer and including a first active portion of the switching element, wherein the source and the drain of the switching element are electrically connected to the first active portion, respectively.

3. The display panel of claim 2, wherein the photosensitive device further includes: a storage capacitor including a first capacitor electrode and a second capacitor electrode at least partially overlapping the first capacitor electrode; the first capacitor electrode is electrically connected to both the first electrode and the second electrode; and the second capacitor electrode is electrically connected to both the first protection electrode and the second protection electrode, and at least partially overlaps the first capacitor electrode.

4. The display panel of claim 3, further comprising: a fourth conductive layer including a conductive portion, wherein an orthographic projection of the conductive portion on the substrate at least partially overlaps an orthographic projection of at least one of the first electrode and the second electrode on the substrate.

5. The display panel of claim 3, wherein the metal oxide semiconductor of the photosensitive layer has an electron mobility greater than or equal to $10 \text{ cm}^2/\text{Vs}$, and the insulating layer has a thickness of 5 nm to 15 nm.

6. The display panel of claim 2, further comprising: a pixel driving circuit including both a first driving transistor and a second driving transistor; wherein the display panel further comprises: a second active layer disposed between the substrate and the photosensitive layer and including a second active portion of the first driving transistor, wherein the second active portion includes a polysilicon semiconductor; a third active layer disposed between the second active layer and the photosensitive layer and including a third active portion of the second driving transistor, wherein the third active portion includes a metal oxide semiconductor; and a conductive layer disposed between the second active layer and the third active layer and including a second gate of the first driving transistor, wherein the second gate at least partially overlaps the second active portion; wherein the second conductive layer further includes a third gate of the second driving transistor, the third gate at least partially overlapping the third active portion.

7. The display panel of claim 6, wherein the first active portion includes a polysilicon semiconductor; and the second active layer further includes the first active portion, and the conductive layer further includes a first gate of the switching element, the first gate at least partially overlapping the first active portion.

8. The display panel of claim 6, wherein the first active portion includes a metal oxide semiconductor; and the third active layer further includes the first active portion, the conductive layer further includes a fourth gate of the switching element, and the second conductive layer further includes a first gate of the switching element, wherein the fourth gate at least partially

overlaps the first active portion, and the first gate at least partially overlaps the first active portion.

9. The display panel of claim 1, further comprising: a plurality of light emitting units arranged in a plurality of rows in a first direction and arranged in a plurality of columns in a second direction; wherein a plurality of intersecting regions are formed between the plurality of rows of light emitting units and the plurality of columns of light emitting units, and each of the intersecting regions is provided with at most one first photosensitive element or one second photosensitive elements.

10. The display panel of claim 9, wherein the first photosensitive element and the second photosensitive element are respectively located in two adjacent ones of the intersecting regions located on the same side of the same light emitting unit.

11. The display panel of claim 10, further comprising: a plurality of the photosensitive devices; wherein a plurality of the first photosensitive elements and a plurality of the second photosensitive elements in the same row are alternately arranged, a plurality of the first photosensitive elements and a plurality of the second photosensitive elements in the same column are alternately arranged, and each of the first photosensitive elements is connected in parallel with one of the second photosensitive elements, which is located in the same row and in the same side as the first photosensitive element and adjacent to the first photosensitive element, or each of the first photosensitive elements is connected in parallel with one of the second photosensitive elements, which is located in the same column and in the same side as the first photosensitive element and adjacent to the first photosensitive element; or a plurality of the first photosensitive elements and a plurality of the second photosensitive elements in the same row are alternately arranged, the first photosensitive elements or the second photosensitive elements are in the same column, and each of the first photosensitive elements is connected in parallel with one of the second photosensitive elements, which is located in the same row and in the same side as the first photosensitive element and adjacent to the first photosensitive element.

12. The display panel of claim 9, wherein the photosensitive device further includes a third photosensitive element; wherein the third photosensitive element includes a third electrode, a third protection electrode, and a third photosensitive portion between the third electrode and the third protection electrode; the third electrode and the first electrode are located in the same layer and connected to each other, the third photosensitive portion and the first photosensitive portion are located in the same layer and arranged at intervals, and the third protection electrode is connected to either the source or the drain; and the first photosensitive element, the second photosensitive element, and the third photosensitive element are respectively located in three of the intersection regions at three vertices of the same light emitting unit.

13. A display panel, comprising: at least one photosensitive device, the at least one photosensitive device including a switching element, a first photosensitive element, and a second photosensitive element; wherein the display panel further comprises: a substrate; a first conductive layer disposed on the substrate and including a first electrode and a second electrode connected to the first electrode; a photosensitive layer disposed on one side of the first conductive layer away from the substrate and including a first photosensitive portion corresponding to the first electrode and a second photosensitive portion corresponding to the second electrode, wherein the first photosensitive portion and the second photosensitive portion are disposed at intervals, and the photosensitive layer further includes a metal oxide semiconductor; an insulating layer disposed between the first conductive layer and the photosensitive layer; a second conductive layer disposed on one side of the photosensitive layer away from the substrate and including a first protection electrode and a second protection electrode, wherein the first protection electrode is electrically connected to the first photosensitive portion, and the second protection electrode is electrically connected to the second photosensitive portion; and a third conductive layer disposed on one side of the second conductive layer away from the substrate and including a source and a drain of the switching element, wherein the first protection electrode and the second protection electrode are

both electrically connected to the source or the drain of the switching element; wherein the first photosensitive element includes the first electrode, the first photosensitive portion, and the first protection electrode, and the second photosensitive element includes the second electrode, the second photosensitive portion, and the second protective electrode; wherein the display panel further comprises: a first active layer disposed between the substrate and the first conductive layer and including a first active portion of the switching element, wherein the source and the drain of the switching element are electrically connected to the first active portion; wherein the display panel further comprises: a plurality of light emitting units arranged in a plurality of rows in a first direction and arranged in a plurality of columns in a second direction; wherein a plurality of intersecting regions are formed between the plurality of rows of light emitting units and the plurality of columns of light emitting units, and each of the intersecting regions is provided with at most one first photosensitive element or one second photosensitive elements.

14. The display panel of claim 13, wherein the photosensitive device further includes: a storage capacitor including a first capacitor electrode and a second capacitor electrode at least partially overlapping the first capacitor electrode; the first capacitor electrode is electrically connected to both the first electrode and the second electrode; and the second capacitor electrode is electrically connected to both the first protection electrode and the second protection electrode, and at least partially overlapping the first capacitor electrode.

15. The display panel of claim 14, further comprising: a fourth conductive layer including a conductive portion, wherein an orthographic projection of the conductive portion on the substrate at least partially overlaps an orthographic projection of at least one of the first electrode and the second electrode on the substrate.

16. The display panel of claim 14, wherein the metal oxide semiconductor of the photosensitive layer has an electron mobility greater than or equal to $10 \text{ cm}^2/\text{Vs}$, and the insulating layer has a thickness of 5 nm to 15 nm.

17. The display panel of claim 13, further comprising: a pixel driving circuit including both a first driving transistor and a second driving transistor; wherein the display panel further comprises: a second active layer disposed between the substrate and the photosensitive layer and including a second active portion of the first driving transistor, wherein the second active portion includes a polysilicon semiconductor; a third active layer disposed between the second active layer and the photosensitive layer and including a third active portion of the second driving transistor, wherein the third active portion includes a metal oxide semiconductor; and a conductive layer disposed between the second active layer and the third active layer and including a second gate of the first driving transistor, wherein the second gate at least partially overlaps the second active portion; wherein the second conductive layer further includes a third gate of the second driving transistor, the third gate at least partially overlapping the third active portion.

18. The display panel of claim 17, wherein the first active portion includes a polysilicon semiconductor; and the second active layer further includes the first active portion, and the conductive layer further includes a first gate of the switching element, the first gate at least partially overlapping the first active portion.

19. The display panel of claim 17, wherein the first active portion includes a metal oxide semiconductor; and the third active layer further includes the first active portion, the conductive layer further includes a fourth gate of the switching element, and the second conductive layer further includes a first gate of the switching element, wherein the fourth gate at least partially overlaps the first active portion, and the first gate at least partially overlaps the first active portion.

20. The display panel of claim 13, wherein the first photosensitive element and the second photosensitive element are respectively located in two adjacent ones of the intersecting regions located on the same side of the same light emitting unit.
